INFO6002 Database Management 2
Assignment 1

Task 1: Conceptual Design

Group Member Names:

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Lab Timings: **Thursday 1.00pm to 3.00pm**

Course coordinator: **Prof. Mahakim Newton**

DATA REQUIREMENTS

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a. Business Organisation Entities:

1. Branch:

Each branch represents a physical location where GlobalCarHire operates its car rental services. Branches across Australia (Sydney, Melbourne, Brisbane, Perth, Adelaide) and internationally (New Zealand, Singapore, Thailand, Indonesia, and Malaysia).

Attributes: branchId, name, address, country, phoneNumber, currency, operationalStatus.

Note: Operational status can be Active, Inactive, or Closed

2. Vehicle:

Each vehicle represents a car owned by a branch and available for rental.

Attributes: vehicleId, make, model, year, registrationNumber, status, mileage, seatingCapacity, fuelType, transmissionType, baseRentalRate

Note:

Status: Active, Inactive, Decommissioned (Temporarily not in service or permanently removed from service).

Fuel Type: Diesel, Petrol, Electric

Transmission Type: Manual, Automatic

3. Vehicle Category:

Each vehicle belongs to a particular vehicle category based on their type and features.

Attributes: categoryId, name, description, dailyRateMultiplier

Note: Vehicle Categories include Economy, Standard, Luxury, and Commercial

4. Employee:

Staff members working at various branches with different authority levels.

Attributes: employeeId, name, position

Note: Position can be Manager or Associate

5. Customer:

Individuals who rent vehicles from GlobalCarHire.

Attributes: customerId, fullName, contactDetails, email, licenseInformation, customerHistory, loyaltyTier

Note: Loyalty tiers include Silver, Gold, Platinum which offers benefits such as discounts.

b. Service Management Entities:

6. Service:

Base entity for all services offered by GlobalCarHire.

Attributes: serviceId, name, description, status

Note: Status can be Available or Unavailable

7. Regular Service:

Standard services offered by GlobalCarHire with consistent pricing.

Attributes: basePrice, currency, restrictions

Note: Regular Service Includes Self-Drive, Chauffeur-Driven, Airport-Shuttle, Fleet-Leasing

8. Package Service:

Bundled services with special pricing and conditions.

Attributes: basePrice, currency, validPeriodStart, validPeriodEnd, inclusions, exclusions, gracePeriod(for cancellations).

Note: Examples include Weekend packages, corporate packages.

9. Priced Service:

Advertised pricing for services which is regular or package by specific branches.

Attributes: pricedServiceId, price, pricingStartDate, pricingEndDate, approverManagerId

Note: A priced package service's start and end date must be within the package's valid period.

10. Extra Service:

Additional services that can be added during rental period.

Attributes: extraServiceId, name, description, advertisedRate

Note: Extra Service Includes child seats, navigators, additional drivers, insurance upgrades

c. Booking and Transaction Entities:

11. Booking:

Core entity representing a customer's booking for vehicle rental services. Customers can book priced services through the company website, app, phone, or in person.

Attributes: bookingId, modeOfBooking, bookingDate, customerInformation, numberOfPricedServices, numberOfVehicles, serviceStartDate, serviceEndDate, paymentMethod, bookingDeposit, bookingStatus.

Note: Status can be Confirmed, Cancelled, or Serviced

12. Driver:

Individuals authorized to drive the rented vehicles.

Attributes: driverId, name, licenseInformation, contactNumbers

13. Vehicle Pickup:

Records maintained for the actual pickup of vehicles by customers.

Attributes: pickupId, actualPickupTime, vehicleConditionPickup

14. Vehicle Return:

Records the return of vehicles and associated charges.

Attributes: returnId, dropOffTime, vehicleConditionReturn, latePenalty, damagePenalty

15. Invoice:

Final billing document including all charges and payments.

Attributes: invoiceId, baseRentalCharges, extraServiceCharges, totalLatePenalty, totalDamagePenalty, totalDiscount, finalAmount, paymentMethod.

TRANSACTION REQUIREMENTS

a. Data Manipulations:

- **Insert/update/delete Branch information**
  - Add new branches with all required details (ID, name, address, status)
  - Modify existing branch details (address, phone, status)
  - Remove branches (with validation for active bookings/vehicles)

- Insert/update/delete Vehicle details and status
  - Register new vehicles with specifications (make, model, category)
  - Update vehicle attributes (mileage, status, rental rate)
  - Decommission vehicles (status change to "Decommissioned")

- Insert/update/delete Vehicle Categories
  - Define new categories (Economy, Standard, Luxury, Commercial.)
  - Modify category descriptions and rate multipliers
  - Remove obsolete categories (with vehicle reassignment)

- Insert/update/delete Employee records
  - Hire new employees with position assignments
  - Update employee details or transfer between branches
  - Terminate employee records (with activity auditing)

- Insert/update/delete Customer information
  - Create new customer profiles with license details
  - Update contact information and loyalty status
  - Archive inactive customer records

- Insert/update/delete Service offerings (Regular and Package)
  - Add new services (Self-Drive, Chauffeur, etc.)
  - Modify service descriptions and restrictions
  - Discontinue services (with booking validation)

- Insert/update/delete Priced Services
  - Set time-bound pricing for services at branches
  - Adjust pricing within validity periods
  - Remove outdated pricing schemes

- Insert/update/delete Booking transactions
  - Create new bookings with service/vehicle assignments
  - Modify booking details before service start
  - Cancel bookings (with penalty calculations)

- Insert/update/delete Driver information
  - Add additional drivers to bookings
  - Update driver license/contact details
  - Remove driver associations

- Insert/update/delete Vehicle pickup and return records
  - Record pickup time/vehicle condition
  - Update return details (time, mileage, condition)
  - Correct erroneous pickup/return entries

- Insert/update/delete Extra services and charges
  - Offer new add-on services (child seats, GPS, etc.)
  - Adjust pricing for extra services
  - Discontinue unavailable extras

- Insert/update/delete Invoice records
  - Generate initial invoices
  - Adjust charges (damages, late fees, discounts)
  - Void/correct erroneous invoices

Queries:

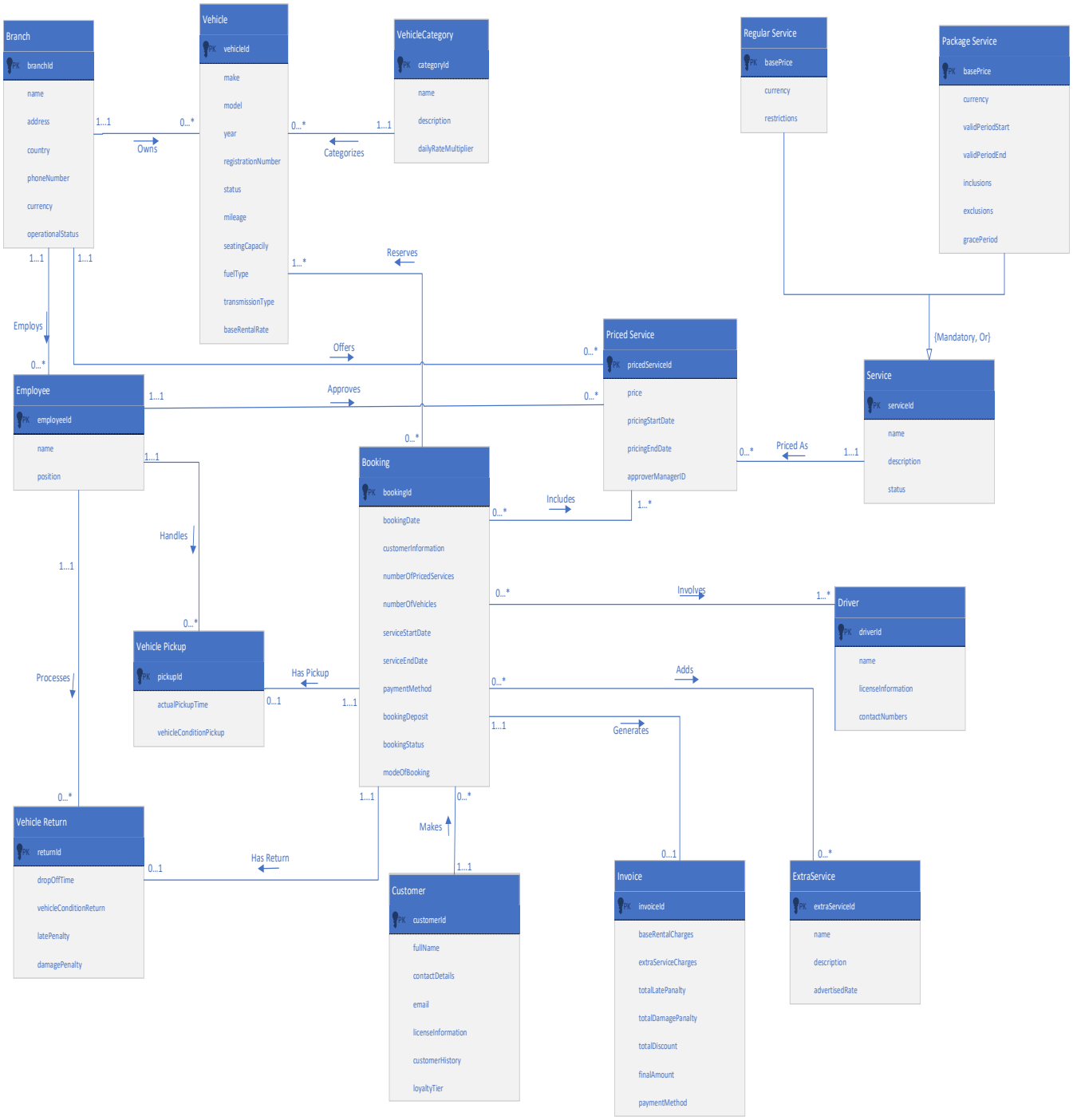
- Search for available vehicles by branch, category, and date range.
- List all active branches by country and operational status.
- Find available priced services for specific date ranges.
- Generate reports of bookings by branch, date, and status.
- List all vehicles due for return on specific dates.
- Find customers with overdue rentals.
- Generate revenue reports by branch and time period.
- List all cancelled bookings with penalties.
- Find vehicles requiring maintenance based on mileage.
- Generate customer loyalty tier reports.
- List all package services with expiring validity periods.
- Find all employees by branch and position.
- Generate invoice summaries by date range.
- List all extra services utilized during specific periods.

Business Rules:

- Each branch must have a unique branchId.
- Operational status must be Active, Inactive, or Closed.
- Each branch must specify its local currency.

- Only Active branches can process new bookings.
- Each vehicle must have a unique vehicleId and registrationNumber.
- Vehicle status must be Active, Inactive, or Decommissioned.
- Only Active vehicles can be included in bookings.
- Each vehicle must belong to exactly one vehicle category.
- Fuel type must be Diesel, Petrol, or Electric.
- Transmission type must be Manual or Automatic.
- Service status must be Available or Unavailable.
- Chauffeur-Driven services are only offered with Luxury category vehicles.
- Fleet-Leasing services are restricted to Commercial category vehicles.
- Package service start and end dates must be within the valid period.
- Priced service dates must be within package validity for package services.
- Each booking must have a unique bookingId.
- Service dates must be within priced service availability dates
- Vehicles cannot be double-booked for overlapping periods.
- Booking status must be Confirmed, Cancelled, or Serviced.
- Cancellation after grace period incurs 25% penalty for package services.
- Each booking may involve multiple drivers.
- Each customer must have valid license information.
- Loyalty tiers (Silver, Gold, Platinum) provide discount benefits.
- Associates can only give discounts to loyal customers.
- Manager approval required for discounts to non-loyal customers.
- Employee position must be Manager or Associate.
- All pricing must use the branch's local currency.
- Late returns incur additional charges.
- Vehicle damage incurs penalty charges.
- Final amount considers all charges, discounts, and deposits.

EER DIAGRAM


Branch
branchId
name
address
country
phoneNumber
currency
operationalStatus
Vehicle
vehicleId
make
model
year
registrationNumber
status
mileage
seatingCapacity
fuelType
transmissionType
baseRentalRate
VehicleCategory
categoryId
name
description
dailyRateMultiplier
Regular Service
basePrice
currency
restrictions
Package Service
basePrice
currency
validPeriodStart
validPeriodEnd
inclusions
exclusions
gracePeriod
Owns
Categorizes
Reserves
Employs
Offers
{Mandatory, Or}
Priced Service
pricedServiceId
price
pricingStartDate
pricingEndDate
approverManagerID
Service
serviceId
name
description
status
Priced As
Employee
employeeId
name
position
Approves
Handles
Booking
bookingId
bookingDate
customerInformation
numberOfPricedServices
numberOfVehicles
serviceStartDate
serviceEndDate
paymentMethod
bookingDeposit
bookingStatus
modeOfBooking
Includes
Involves
Driver
driverId
name
licenseInformation
contactNumbers
Vehicle Pickup
pickupId
actualPickupTime
vehicleConditionPickup
Has Pickup
Processes
Adds
Generates
Vehicle Return
returnId
dropOffTime
vehicleConditionReturn
latePenalty
damagePenalty
Has Return
Makes
Customer
customerId
fullName
contactDetails
email
licenseInformation
customerHistory
loyaltyTier
Invoice
invoiceId
baseRentalCharges
extraServiceCharges
totalLatePanalty
totalDamagePanalty
totalDiscount
finalAmount
paymentMethod
ExtraService
extraServiceId
name
description
advertisedRate

DATA DICTIONARY

a) Entity Type Table:

ENTITY	DESCRIPTION	ALIASES	OCCURRENCES
Branch	Physical location where GlobalCarHire operates	Location, Office	Multiple branches across countries
Vehicle	Cars owned by branches for rental	Car, Automobile	Fleet of vehicles per branch
Vehicle Category	Classification of vehicles by type and features	Car Type, Vehicle Class	Economy, Standard, Luxury, Commercial
Employee	Staff members working at branches	Staff, Worker	Managers and Associates at each branch
Customer	Individuals who rent vehicles	Client, Renter	Registered users of the system
Service	Base entity for all services offered	Offering, Product	Various rental services
Regular Service	Standard services with consistent pricing	Standard Service	Subclass of Service
Package Service	Bundled services with special conditions	Bundle, Deal	Subclass of Service
Priced Service	Advertised pricing for services by branches	Service Pricing	Branch-specific service pricing
Booking	Customer reservation for vehicle rental	Reservation, Order	Customer bookings for services
Driver	Authorized drivers for rented vehicles	Authorized Driver	Multiple drivers per booking allowed
Vehicle Pickup	Record of vehicle collection by customer	Collection, Handover	One pickup per booking

Vehicle Return	Record of vehicle return and condition	Drop-off, Return	One return per booking
Extra Service	Additional services added during rental	Add-on, Enhancement	Optional services for bookings
Invoice	Final billing document with all charges	Bill, Receipt	Generated upon vehicle return

b) Relationship Type Table:

ENTITY NAME	MULTIPLICITY	RELATIONSHIP	MULTIPLICITY	ENTITY NAME
Branch	1..1	Owns	0..*	Vehicle
Vehicle Category	1..1	Categorizes	0..*	Vehicle
Branch	1..1	Employs	0..*	Employee
Service	{Man, OR}	ISA	{Man, OR}	Regular Service
Service	{Man, OR}	ISA	{Man, OR}	Package Service
Service	1..1	PricedAs	0..*	Priced Service
Branch	1..1	Offers	0..*	Priced Service
Employee	1..1	Approves	0..*	Priced Service
Customer	1..1	Makes	0..*	Booking
Booking	0..*	Includes	1..*	Priced Service
Booking	0..*	Reserves	1..*	Vehicle
Booking	0..*	Involves	1..*	Driver
Booking	1..1	HasPickup	0..1	Vehicle Pickup
Booking	1..1	HasReturn	0..1	Vehicle Return
Booking	1..1	Generates	0..1	Invoice
Employee	1..1	Handles	0..*	Vehicle Pickup
Employee	1..1	Processes	0..*	Vehicle Return
Booking	0..*	Adds	0..*	Extra Service

c) Attribute Type Table:

ENTITY	ATTRIBUTES	DESCRIPTION	DATA TYPE & LENGTH	NULLS	MULTIVALUED	DERIVED	DEFAULT
Branch	branchId	Unique identifier for the branch	VARCHAR(10)	NO	NO	NO	-
	name	Name of the branch	VARCHAR(50)	NO	NO	NO	-
	address	Physical address of the branch	VARCHAR(100)	NO	NO	NO	-
	country	Country where the branch is located	VARCHAR(50)	NO	NO	NO	-
	phoneNumber	Contact number of the branch	VARCHAR(15)	NO	NO	NO	-
	currency	Local currency used by the branch	VARCHAR(3)	NO	NO	NO	-
	operationalStatus	Status of the branch (Active/Inactive/Closed)	ENUM	NO	NO	NO	'Active'
Vehicle	vehicleId	Unique identifier for the vehicle	VARCHAR(10)	NO	NO	NO	-
	make	Manufacturer of the vehicle	VARCHAR(20)	NO	NO	NO	-
	model	Model of the vehicle	VARCHAR(20)	NO	NO	NO	-
	year	Manufacturing year of the vehicle	INT	NO	NO	NO	-
	registrationNumber	License plate number of the vehicle	VARCHAR(15)	NO	NO	NO	-

	status	Status of the vehicle (Active/Inactive/Decommissioned)	ENUM	NO	NO	NO	'Active'
	mileage	Current mileage of the vehicle	INT	NO	NO	NO	0
	seatingCapacity	Number of seats in the vehicle	INT	NO	NO	NO	-
	fuelType	Type of fuel (Diesel/Petrol/Electric)	ENUM	NO	NO	NO	-
	transmissionType	Transmission type (Manual/Automatic)	ENUM	NO	NO	NO	-
	baseRentalRate	Base rental rate per day	DECIMAL(10,2)	NO	NO	NO	-
VehicleCategory	categoryId	Unique identifier for the category	VARCHAR(10)	NO	NO	NO	-
	name	Name of the category (Economy/Standard/Luxury/Commercial)	VARCHAR(20)	NO	NO	NO	-
	description	Description of the category	VARCHAR(100)	YES	NO	NO	NULL
	dailyRateMultiplier	Multiplier applied to base rental rate	DECIMAL(3,2)	NO	NO	NO	1
Employee	employeeId	Unique identifier for the employee	VARCHAR(10)	NO	NO	NO	-
	name	Name of the employee	VARCHAR(50)	NO	NO	NO	-
	position	Position of the employee (Manager/Associate)	ENUM	NO	NO	NO	-
Customer	customerId	Unique identifier for the customer	VARCHAR(10)	NO	NO	NO	-
	fullName	Full name of the customer	VARCHAR(50)	NO	NO	NO	-
	contactDetails	Contact details of the customer	VARCHAR(50)	NO	NO	NO	-

	email	Email address of the customer	VARCHAR(50)	NO	NO	NO	-
	licenseInformation	Driver's license details	VARCHAR(50)	NO	NO	NO	-
	customerHistory	Rental history of the customer	TEXT	YES	YES	YES	NULL
	loyaltyTier	Loyalty tier (Silver/Gold/Platinum)	ENUM	NO	NO	NO	'Silver'
Service	serviceId	Unique identifier for the service	VARCHAR(10)	NO	NO	NO	-
	name	Name of the service	VARCHAR(50)	NO	NO	NO	-
	description	Description of the service	VARCHAR(100)	YES	NO	NO	NULL
	status	Status of the service (Available/Unavailable)	ENUM	NO	NO	NO	'Available'
Regular Service	basePrice	Base price for the service	DECIMAL(10,2)	NO	NO	NO	-
	currency	Currency for the service	VARCHAR(3)	NO	NO	NO	-
	restrictions	Restrictions for the service	VARCHAR(100)	YES	NO	NO	NULL
Package Service	basePrice	Base price for the package	DECIMAL(10,2)	NO	NO	NO	-
	currency	Currency for the package	VARCHAR(3)	NO	NO	NO	-
	validPeriodStart	Start date of the package validity	DATE	NO	NO	NO	-
	validPeriodEnd	End date of the package validity	DATE	NO	NO	NO	-
	inclusions	Services included in the package	TEXT	YES	YES	NO	NULL
	exclusions	Services excluded from the package	TEXT	YES	YES	NO	NULL
	gracePeriod	Cancellation grace period (in days)	INT	NO	NO	NO	1

Priced Service	pricedServiceId	Unique identifier for the priced service	VARCHAR(10)	NO	NO	NO	-
	price	Advertised price for the service	DECIMAL(10,2)	NO	NO	NO	-
	pricingStartDate	Start date of the pricing period	DATE	NO	NO	NO	-
	pricingEndDate	End date of the pricing period	DATE	NO	NO	NO	-
	approverManagerId	ID of the manager who approved the pricing	VARCHAR(10)	YES	NO	NO	NULL
Extra Service	extraServiceId	Unique identifier for the extra service	VARCHAR(10)	NO	NO	NO	-
	name	Name of the extra service	VARCHAR(50)	NO	NO	NO	-
	description	Description of the extra service	VARCHAR(100)	YES	NO	NO	NULL
	advertisedRate	Rate for the extra service	DECIMAL(10,2)	NO	NO	NO	-
Booking	bookingId	Unique identifier for the booking	VARCHAR(10)	NO	NO	NO	-
	modeOfBooking	Method used to make the booking (Website/App/Phone/In-person)	ENUM	NO	NO	NO	-
	bookingDate	Date when the booking was made	DATETIME	NO	NO	NO	CURRENT_DATE
	customerInformation	Details of the customer making the booking	VARCHAR(10)	NO	NO	NO	-
	numberOfPricedServices	Number of priced services included	INT	NO	NO	NO	1
	numberOfVehicles	Number of vehicles reserved	INT	NO	NO	NO	1
	serviceStartDate	Start date of the service period	DATE	NO	NO	NO	-

	serviceEndDate	End date of the service period	DATE	NO	NO	NO	-
	paymentMethod	Method of payment for the booking	ENUM	NO	NO	NO	-
	bookingDeposit	Deposit amount paid for the booking	DECIMAL(10,2)	YES	NO	NO	0
	bookingStatus	Status of the booking (Confirmed/Cancelled/Serviced)	ENUM	NO	NO	NO	'Confirmed'
Driver	driverId	Unique identifier for the driver	VARCHAR(10)	NO	NO	NO	-
	name	Name of the driver	VARCHAR(50)	NO	NO	NO	-
	licenseInformation	Driver's license details	VARCHAR(50)	NO	NO	NO	-
	contactNumbers	Contact numbers of the driver	VARCHAR(50)	YES	YES	NO	NULL
Vehicle Pickup	pickupId	Unique identifier for the pickup record	VARCHAR(10)	NO	NO	NO	-
	actualPickupTime	Actual time when the vehicle was picked up	DATETIME	NO	NO	NO	-
	vehicleConditionPickup	Condition of the vehicle at pickup	TEXT	NO	NO	NO	-
Vehicle Return	returnId	Unique identifier for the return record	VARCHAR(10)	NO	NO	NO	-
	dropOffTime	Time when the vehicle was returned	DATETIME	NO	NO	NO	-
	vehicleConditionReturn	Condition of the vehicle at return	TEXT	NO	NO	NO	-
	latePenalty	Penalty for late return	DECIMAL(10,2)	YES	NO	NO	0
	damagePenalty	Penalty for vehicle damage	DECIMAL(10,2)	YES	NO	NO	0
Invoice	invoiceId	Unique identifier for the invoice	VARCHAR(10)	NO	NO	NO	-

	baseRentalCharges	Charges for the base rental	DECIMAL(10,2)	NO	NO	NO	-
	extraServiceCharges	Charges for extra services	DECIMAL(10,2)	YES	NO	NO	0
	totalLatePenalty	Total penalty for late return	DECIMAL(10,2)	YES	NO	NO	0
	totalDamagePenalty	Total penalty for vehicle damage	DECIMAL(10,2)	YES	NO	NO	0
	totalDiscount	Total discount applied	DECIMAL(10,2)	YES	NO	NO	0
	finalAmount	Final amount to be paid	DECIMAL(10,2)	NO	NO	YES	-
	paymentMethod	Method of payment for the invoice	ENUM	NO	NO	NO	-